

Final Project Submission: Clustering NBA Players

1. Introduction

In professional basketball, teams are constantly looking for ways to gain a competitive edge, especially in talent acquisition and player development.

Scouting has traditionally relied on observational techniques, interviews, and simple statistics like points per game, but this usually leaves talented players overlooked. These players might not get much court time but show performance characteristics that align with top-tier talent.

My project addresses this inefficiency through a data-driven approach.

The goal is to uncover 'Hidden Gems'—NBA players whose advanced performance metrics are comparable to elite players, but who do not receive high playing time.

To do this, I use unsupervised machine learning (KMeans clustering) to identify playing styles and roles beyond traditional positions.

Then I built a supervised model to predict players who fit the Hidden Gem profile.

This analysis leverages multiple player metrics such as scoring, rebounding, assists, efficiency, and defensive stats.

This type of analysis can offer great value to stakeholders like coaches, general managers, and analysts.

For example, a team with limited budget space can identify high-value players before they ask for large contracts.

Similarly, a developmental coach can tailor training to suit a player's hidden strengths.

In a market that values finding the next breakout star, data can help drive smarter, faster decisions.

To gather relevant data, I used a dataset from Kaggle titled 'NBA Player Stats 2022–2023 - Regular Season.'

It has metrics for every NBA player including minutes played, points per game, shooting percentages, rebounds, assists, steals, and more.

These metrics were selected because they represent a comprehensive snapshot of player contribution across multiple dimensions.

2. Milestone 1: Exploratory Data Analysis

The first milestone in the project focused on understanding the structure and distribution of the dataset.

Exploratory Data Analysis (EDA) was performed using descriptive statistics, correlation matrices, and visualizations.

Histograms and box plots were used to observe the distribution of key features such as points, minutes played, field goal percentage, and rebounds.

This helped identify trends such as a small percentage of players dominating in scoring and playing time, which aligns with real-world team strategies.

To uncover patterns in the dataset, I used Principal Component Analysis (PCA) to reduce the dimensionality while preserving most of the variance.

This allowed for more effective clustering and visualization.

I then applied the KMeans clustering algorithm to segment players based on playing style.

Scatter plots of the PCA results, color-coded by cluster, helped visually separate player types such as scorers, defenders, facilitators, or all-around contributors.

I also analyzed how clusters relate to traditional player positions using stacked bar plots. Some clusters were dominated by point guards or centers, but others had a more diverse composition showing that the model was identifying roles and tendencies that go beyond standard positional classifications.

This finding supports the hypothesis that modern basketball is becoming positionless, and data-driven segmentation is more insightful.

3. Milestone 2: Data Preparation

With insights gained from the EDA phase, I moved on to preparing the dataset for modeling.

This included selecting a subset of features that represented core contributions from players, such as points, assists, rebounds, steals, and shooting percentages.

The data was cleaned by handling missing values and standardizing the features to make sure that there was uniform scale across all metrics, which is important for both clustering and classification.

One important feature created was the cluster label, derived from the KMeans results. This feature groups players based on similar statistical profiles.

Another important engineered feature was the 'Hidden_Gem' label.

Players were marked as Hidden Gems if they fell below a threshold for average minutes played but above thresholds for key performance metrics.

The thresholds were chosen based on quantiles to ensure a balance of player representation.

This label served as the target for the classification model and represents players who are currently underutilized but show traits associated with elite performance.

Adding this label adds strategic value by identifying overlooked potential and preparing the dataset for downstream machine learning tasks.

4. Milestone 3: Model Building and Evaluation

The third milestone focused on building and evaluating a predictive model to classify players as Hidden Gems or not.

I chose the Random Forest Classifier for its ability to handle nonlinear data, and built-in feature importance analysis.

The dataset was split into training and testing sets to evaluate the model's performance on unseen data.

Key metrics for evaluation included accuracy (how often the model is right), precision (how many of the predicted Hidden Gems were truly Hidden Gems), recall (how many of the actual Hidden Gems were correctly predicted), and F1-score (a balance between precision and recall).

The results showed high scores across all metrics, with a confusion matrix confirming that most players were correctly classified.

However, the perfect scores on test data suggest potential overfitting, and additional validation would be needed before deployment.

A classification report and confusion matrix were included in the notebook to interpret results visually.

These helped identify which types of players the model was most successful in detecting and where it might require improvement.

5. Conclusion and Recommendations

The project demonstrated that it's possible to use performance data to identify NBA players who may be undervalued or overlooked.

By combining clustering and classification techniques, the project produced players whose statistical profiles align with star-caliber players despite limited playing time.

The model is not yet deployment-ready. Even though its performance is strong, it needs to be tested on different seasons and possibly expanded to other leagues like the NCAA or G-League.

Also, more sophisticated modeling techniques, such as XGBoost or neural networks, could be explored to refine performance.

My recommendations include the following:

- Expand the dataset to include multiple seasons and external sources.
- Incorporate contextual variables like salary, age, contract status, or team role.
- Develop a user-facing scouting dashboard that visualizes key model insights.
- Explore transferability of this model to other sports or talent identification problems.

In conclusion, this project lays a strong foundation for data-driven scouting and strategic decision-making.

As basketball continues to evolve, using analytics to uncover potential early will be key to gaining a long-term advantage.

Updated Model Evaluation Insights

After balancing the training data using SMOTE and constructing a more balanced test set, the model's ability to correctly identify Hidden Gems improved significantly.

In the updated evaluation:

- The model identified both actual Hidden Gems in the test set.
- Precision for Hidden Gems improved to 67%, meaning two-thirds of the predicted Hidden Gems were actually correct.
- Recall for Hidden Gems reached 100%, capturing all of the real ones.
- The F1-Score of 0.80 shows a strong balance between precision and recall.
- Overall model accuracy improved to 90%.

These results are more realistic and practical than earlier evaluations, which had extreme class imbalance.

This version demonstrates the model's growing potential to support scouting by identifying under-the-radar players based on their performance data.